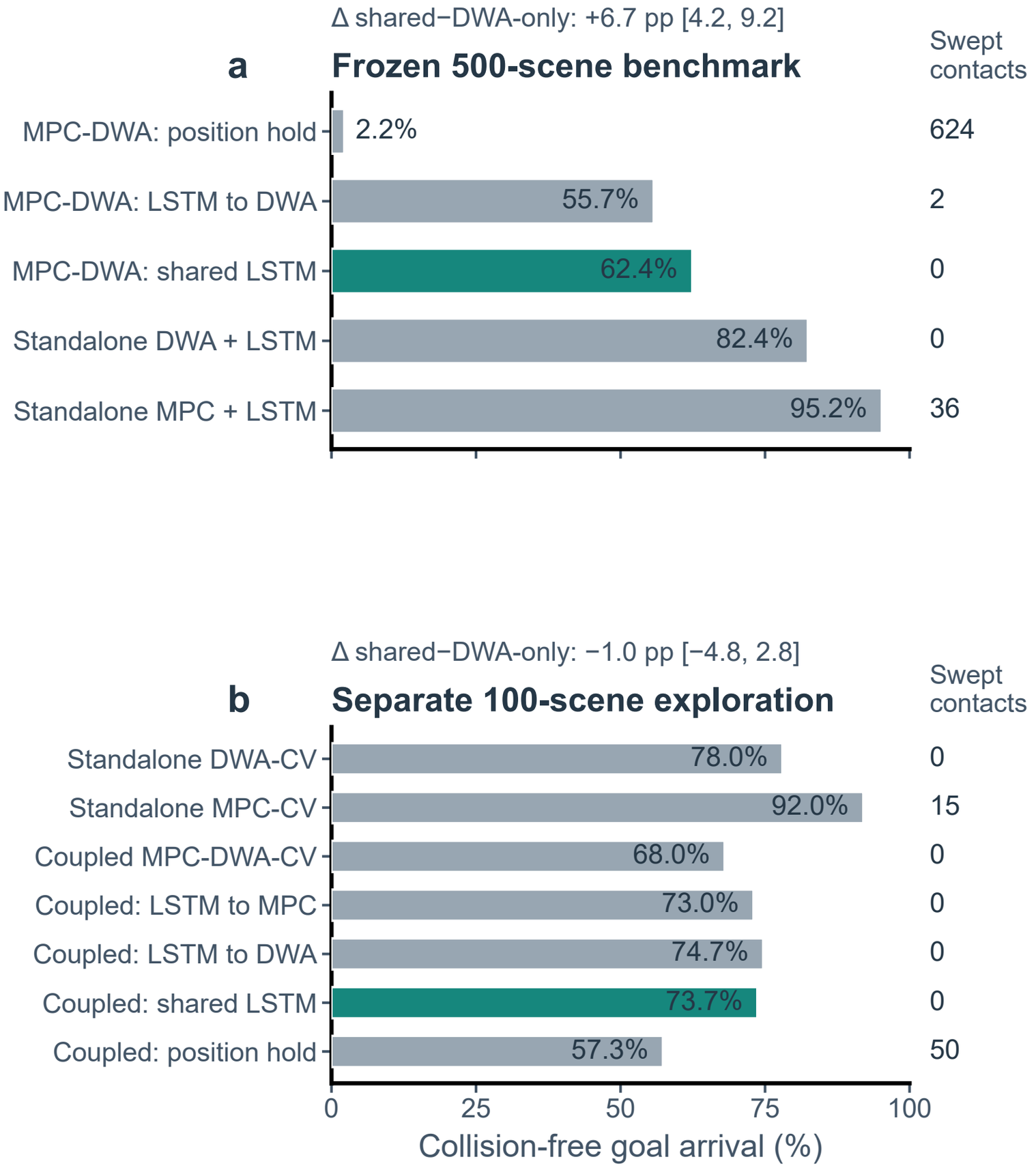
Δ shared−DWA-only: +6.7 pp [4.2, 9.2]
a
Frozen 500-scene benchmark
Swept contacts
MPC-DWA: position hold
2.2%
624
MPC-DWA: LSTM to DWA
55.7%
2
MPC-DWA: shared LSTM
62.4%
0
Standalone DWA + LSTM
82.4%
0
Standalone MPC + LSTM
95.2%
36
Δ shared−DWA-only: −1.0 pp [−4.8, 2.8]
b
Separate 100-scene exploration
Swept contacts
Standalone DWA-CV
78.0%
0
Standalone MPC-CV
92.0%
15
Coupled MPC-DWA-CV
68.0%
0
Coupled: LSTM to MPC
73.0%
0
Coupled: LSTM to DWA
74.7%
0
Coupled: shared LSTM
73.7%
0
Coupled: position hold
57.3%
50
0
25
50
75
100
Collision-free goal arrival (%)